Vyap Patel

Maple Grove, Minnesota, 55369

352-229-1950 vyap.work@gmail.com https://www.linkedin.com/in/vyap-patel

Education

University of Minnesota - Twin Cities **May 2025**

Bachelor of Science in Computer Science (GPA: 3.68) *Minneapolis, MN*

Relevant Coursework:

- Algorithms and Data Structures
- Machine Architecture and Organization
- Operating Systems
- Real Time and Embedded Systems
- Practice Of Database Systems
- Program Design and Development
- Internet Programming
- Introduction to Computer Networks
- Developing Secure Software Systems

Technical Skills

Proficient Languages: Python, Java, SQL, C, C++, JavaScript, HTML, CSS, Assembly, R

Relevant Frameworks/Libraries: React.js, Flask, Express.js, Node.js

Developer Tools: Git, Github, Docker, REST, Doxygen

Projects

De-Bruijn Graph Genome Assembler | *Python, Matplotlib Git, GitHub* **October 2024 - November 2024**

- Led a 4-person development team to architect and implement a scalable genome assembly system using De Bruijn graph algorithms, processing 3+ billion base pair genomes and reducing assembly time by 40%
- Overcame critical memory constraints by designing efficient graph traversal algorithms that reduced RAM usage by 60% when processing large-scale genomic datasets (10+ GB FASTA files)
- Coordinated cross-functional collaboration between computer science and biology team members through technical presentations and code reviews, ensuring project deliverables met academic research deadlines

Fitness Tracker | *C, MPLAB, ATmega3208 AVRBLE, React.js, Node.js, Express.js* **October 2024 - November 2024**

- Led embedded systems development for a comprehensive wearable health monitoring device, integrating 3 critical sensors and reducing power consumption by 35% through optimized interrupt-driven architecture
- Spearheaded cross-platform integration between embedded C firmware and React.js dashboard, mentoring team members on I2C protocol implementation while coordinating real-time BLE data transmission and frontend-backend communication
- Overcame critical timing constraints by implementing sub-100ms response times for real-time health data streaming, solving complex UART GPS communication challenges and establishing robust AVR-BLE connectivity protocols

Drone Delivery Simulation System | *C++, Docker, Git, Github* **February 2024 - May 2024**

- Worked in a team of 4 people using Agile methodologies to build enterprise-scale drone logistics simulation, processing 1000+ delivery scenarios and reducing route optimization time by 38% through advanced C++ algorithms
- Achieved 90% test coverage through comprehensive unit testing, reducing post deployment bugs by 75%
- Implemented comprehensive Git workflow and design patterns while mentoring junior developers on OOP principles and Docker containerization, resulting in 50% faster code review cycles and seamless CI/CD pipeline deployment across development environments

Full Stack Daily Planner | *HTML, CSS, JavaScript, Python, MySQL, Google Maps API, Stocks API* **January 2024 - May 2024**

- Architected end-to-end web application serving 200+ users with personalized scheduling, integrating Google Maps and Stock.js APIs while improving productivity by 30% through real-time data visualization
- Engineered secure full-stack architecture with MySQL database design, SQL authentication protocols, and session management supporting concurrent user access and complex API integrations
- ”Developed intelligent caching strategies reducing API calls by 60% and solving scalability challenges for 10,000+ daily entries while maintaining real-time accuracy

Leadership Experience

Code The Gap — *Curriculum Planner intern* **2024**

- Designed and established Python programming curriculum reaching 200+ K-12 students, achieving 40% increase in female and minority STEM participation
- Created 24+ interactive lesson plans resulting in 85% student engagement rate and 90% assignment completion rate
- Initiated and led bi-weekly feedback sessions with students, resulting in 30% improvement in couse satisfaction scores

Hack Club — *Vice President* **2021**

- Cultivated and guided a 12-member student organization focused on cybersecurity and competitive programming, orchestrating 4 technical workshops serving 50+ participants across the academic year
- Led growth initiatives resulting in 50% membership increase (from 12 to 18)
- Integrated mentorship program pairing experienced members with newcomers, resulting in 50% increase in project completion rates